

Prithwitosh Dey

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Institution: Max Planck Institute for Solar System Research
Göttingen, Germany

Education

PhD studies (Physics) 2022–2026

International Max Planck Research School at the University of Göttingen

Dissertation: “Solar inertial modes as probes of rotation and convection”

Advisors: Prof. Dr. Laurent Gizon

Dr. Damien Fournier

Prof. Dr. Olga Shishkina

Thesis defended on 15th June 2026.

Integrated B.Sc. M. Sc. (Physics) 2017–2022

Center for Excellence in Basic Sciences, Mumbai, India

Distinctions

- Received a travel grant to travel to Armenia for the IAU Symposium No. 365 (2023) .
 - Selected for the Visiting Students’ Research Programme (VSRP) at the Tata Institute for Fundamental Research, Mumbai (2021).
 - Received the Innovation in Science Pursuit for Inspired Research (INSPIRE) Scholarship, awarded by the Department of Science and Technology, Government of India (2017).
 - Received the Jagdish Chandra Bose National Talent Search (JBNSTS) Scholarship (2015).
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Conference Presentations

- Talk titled *Solar inertial modes from eigenvalue solvers* presented at the conference on “Low-Frequency Oscillations of the Sun and Stars” at Berlin in December 2025.
- Talk titled *Sensitivity of solar inertial modes to rotation at high-latitudes* presented at the “SDO 2025 Science Workshop, A Gathering of the Helio-hive!” at Boulder in February 2025.
- Talk titled *Inertial modes in the solar convection zone: The forward problem* presented at the “6th International conference on Mathematical and Numerical Aspects of Wave Propagation” at Berlin in July 2024.
- Talk titled *Probing the superadiabaticity of the solar convection zone with inertial modes* presented at the “IAU Symposium No. 365: Dynamics of Solar and Stellar Convection Zones and Atmospheres” at Yerevan in August 2023.

Refereed publication

Dey, P., Gizon, L., Bekki, Y. (2026), Towards inertial-mode helioseismology: Direct sensing of solar rotation at 75° latitude and $0.8R_\odot$, *Astronomy & Astrophysics*

Paper in preparation

Dey, P. et al. (2026), Probing the superadiabaticity of the solar convection zone with the $m = 1$ high-latitude inertial mode

Draft available at <https://doi.org/10.17617/3.GRE0YI>

Other projects

- A study of helioseismic inversion techniques Autumn 2021
Supervisor: Prof. Jørgen Christensen-Dalsgaard, University of Aarhus
- An analysis of X-ray data from LAXPC Spring 2021
Supervisors: Prof. H. M. Antia, Tata Institute of Fundamental Research
Dr. K. Mukerjee, Tata Institute of Fundamental Research
- Inversions for solar rotation rate using seismic data Autumn 2020
Supervisor: Prof. H. M. Antia, Tata Institute of Fundamental Research

Additional skills

Attended the following courses at the Max Planck Institute for Solar System Research:

Clean code for astrophysicists	Winter 2023
Data Analysis in Astrophysics	Summer 2023
Physics of the Interior of the Sun and Stars	Winter 2022
Solar System Science: The Central Star	Winter 2022

Programming Languages:

Expert: Python, Fortran 90, Bash, LaTeX
Familiar: Mathematica, C, JAVA

Teaching Experience

- Teaching Assistant for the M.Sc course on *Physics of the Interior of the Sun and Stars* at the University of Göttingen (2023)
- Teaching Assistant for the M.Sc course on *Data analysis in Astrophysics* at the University of Göttingen (2024)
- Teaching Assistant for the M.Sc course on *Solar System Science: The Central Star* at the University of Göttingen (2023)

Service

- Member of the PhD Survey Group of the PhDNet of the Max Planck Society
Collected, analysed and interpreted data from PhD students across the Max Planck Society.
Report available at <https://doi.org/10.17617/2.3626858>
- Organized S3 seminars; a weekly internal seminar series at the Max Planck Institute for Solar System Research
- Managed the website of the Solar and Stellar Interiors department at the Max Planck Institute for Solar System Research, <https://www.mps.mpg.de/helioseismology-asteroseismology>
- Organizing Rocks and Stars III: A student organized international conference at the Max Planck Institute for Solar System Research.
- Developed and presented models of acoustic modes at the Night of Science, Göttingen, 2025.